

8. [Maximum mark: 8]

Consider the complex number $z_1 = \sqrt{3} - 3i$.

(a) Express z_1 in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [3]

Consider the complex number $z_2 = 2\sqrt{3}e^{i\frac{5\pi}{6}}$.

The cube roots of $\frac{z_2}{z_1}$ are denoted by u , v and w .

(b) Find u , v and w . [5]

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